# **Industrial Rise of China:** Geography, Big Push, and the Engineered Invisible Hand

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## Abstract

This article identifies geographic volatility, which determines the variance of grain output in pre-modern societies, as a complementary dimension that refines the role of geography in development.

Specifically, building on Jeffrey Sachs's established perspective on geographic endowments, we propose a theoretical framework in which geography determines not only the mean but also the variance of output. The mean is shaped by a broad set of geographic endowments (soil quality, transport access, disease burden), while the variance (volatility) is driven primarily by climatic instability. The mean of output determines the scale of the market; its variance determines the degree of systemic stability. By shaping the time preferences of individual agents and raising enforcement costs, this volatility determines the structural transaction costs of social coordination and steers the trajectory of institutional evolution.

Crucially, while geographic volatility is an inherent natural constraint, modern technology and engineering can stabilize realized output. This framework thus accounts for the Great Divergence, when geographic volatility constrained institutions through volatile food supplies, and explains the heterogeneity of contemporary industrialization, where nations with limited capacity face higher de-risking costs. For late-developing nations, this framework implies that a Big Push is indispensable, but success hinges on whether it can achieve systemic agricultural de-risking.

Ultimately, the geographic volatility perspective reinforces Jeffrey Sachs's emphasis on the interplay of geography, technology, and institutions, strengthening the shared conclusion: geography precedes institutions, and the material foundation must be secured before markets can function.

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## Introduction: The Pivot of History

Today, China's industrial rise is the defining story of the global economy.

Western observers often view this shift simply as a threat to existing power structures or the resurgence of a fundamentally different civilizational model. Yet, these views miss the underlying logic: the central question is not just about geopolitical leadership, but about the principles by which humanity can survive and thrive under different natural constraints. Taking the long view of history, we see that China's path did not start today. It began with the civilization's distinctive foundational challenge: Faced with a land that was both immensely fertile and inherently volatile, how was this civilization to forge order amidst the chaos of nature?

This question points to a gap in economics and the social sciences more broadly. Traditional analysis has focused overwhelmingly on society itself (institutions, culture, policy) while paying insufficient attention to the relationship between humanity and nature. Jeffrey Sachs is a pioneering voice in economics: his work on the interplay of geography, technology, and institutions has forcefully demonstrated that geographic constraints precede institutional outcomes. Within this established perspective, the analysis has focused on the geographic endowments (soil quality, transport access, disease burden) that determine the level of output (the mean). This explains why some regions are poor (low mean), but it struggles to explain why regions with high output, such as the Yangtze Delta, nevertheless failed to industrialize spontaneously.

Building on this foundation, we identify a dimension that has remained underexplored as a determinant of divergent development paths: volatility—the variance of grain output driven by climatic instability. By incorporating volatility (the variance), we aim to offer a more complete account of how nature shapes the trajectory of human development. While the mean of output determines the scale of the market, its variance determines the degree of systemic stability. By shaping the time preferences of individual agents and enforcement costs, this volatility determines the structural transaction costs of social coordination and steers the trajectory of institutional evolution. [1]

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[1] What we term "time preferences shaped by material conditions" corresponds to what Douglass North called "beliefs" or "mental models" in his later work. North treated beliefs as culturally inherited priors that drive institutional choice; we treat them as rational responses to the physical environment, outputs of material conditions, not inputs to institutional design. The Dujiangyan case is dispositive: where ancient irrigation engineering virtually eliminated output variance, agents inside the stability zone, within the same culture, legal tradition, and political system, behaved like English tenant farmers, while those outside it behaved like classic risk-averse peasants. What changed was not belief but variance. This reframing also resolves two further Northian concepts. His "credible commitment" (North and Weingast, 1989), the willingness to honor long-term obligations, presupposes the stable surplus that low variance provides; it is the economic expression of what Mencius called "a steady mind born of a secure livelihood." His "path dependence" is better understood as the persistence of the material conditions of surplus: institutions endure not because they are locked in, but because the conditions that selected them endure. Once those conditions are altered, as China's agricultural engineering demonstrated, institutions shift accordingly.

Crucially, while geographic volatility is an inherent natural constraint, modern technology and engineering can stabilize realized output. This distinction between the natural constraint and the engineered outcome provides the key to understanding both the Great Divergence and the heterogeneity of contemporary industrialization. In short, where England's invisible hand rested on providential stability, China's had to be engineered.

The article first develops the theoretical framework, showing how geographic volatility shapes rational choice, constrains the division of labor, and sets the material threshold of industrialization. It then applies this framework to the historical record—from England's providential stability and spontaneous industrialization, through traditional China's stagnation trap and contemporary China's engineered breakthrough, to the Soviet Union's cautionary failure. Ireland, where stability was decoupled from abundance, provides a supplementary test of the framework's explanatory power. The article concludes with policy implications for the Global South and a reflection on the renewal of civilization.

# Theoretical Framework: Complementing Endowments with Volatility

## 1. The Mean-Variance Framework

- **Geographic endowments (the mean).**
  - The established perspective attributes the failure of tropical or landlocked nations to low output or high costs (e.g., poor soil, expensive transport, high disease burden). This is a mean-based analysis, which accurately captures the development obstacles faced by many regions with harsh geographic conditions.
  - Theoretical limits: while this perspective effectively explains the poverty trap (e.g., in Sub-Saharan Africa), it struggles to explain the “involution trap” (the deadlock of self-sufficiency) as seen in traditional China. China's core regions possessed fertile soil and high yields. Contrary to assumptions of high transport costs, the Yangtze River system provided an extensive inland water transport network comparable to major European systems. Furthermore, located in the temperate and subtropical zones, it supported the world's largest population, indicating a low disease burden. Yet, this region nevertheless fell into stagnation.
- **Geographic volatility (the variance).**
  - The challenge for traditional China, under the continental East Asian monsoon, was not an inability to generate surplus, but the volatility that eroded it and destabilized the entire system built upon that surplus.
  - The destructive power of variance: frequent floods and droughts (high variance) mean that the process of capital accumulation is periodically interrupted or even reversed (a system-wide "reset"). The surplus created in bountiful years is often consumed or destroyed in lean years, locking society into a cycle of accumulation and dissipation. Beyond this direct destruction of capital, high variance also disrupts price signals, shortens planning horizons, and raises enforcement costs, mechanisms we develop in the sections that follow. The core obstacle to industrialization lies not in the initial stock of capital, but in the stability required for capital to flow into long-term industrial investment.

## 2. Micro-Mechanism: From Geographic Volatility to Rational Choice

Geography does not dictate institutions directly; it shapes them by shaping the time preferences of individual agents, determining whether a society organizes around development or around survival.

- **Stable environment (e.g., England):**
  - Price Signaling: in this environment, geographic stability acts as a natural insurer, ensuring that the price system provides clear and reliable signals of value.
  - Rationality: this predictability lowers the discount rate, making a long-term orientation the rational default.
  - Agents’ Behavior: agents dare to specialize, trusting the market to supply what

they no longer produce for themselves.

  - Capital Behavior: capital flows into long-horizon, high-fixed-cost investments (e.g., tenant farming, factories, machinery) because a stable price environment ensures that residual surplus can be effectively captured.
- **Highly volatile environment (e.g., Traditional China):**
  - Price Signaling: here, frequent floods and droughts translate directly into persistent market noise and erratic price signals, rendering the entire price system unreliable for valuation.
  - Rationality: this unreliability forces a high discount rate. Recurrent subsistence crises do not merely raise the required risk premium; they make near-term survival the overriding priority, rendering long-horizon payoffs effectively irrelevant. Short-term orientation becomes the only rational choice.
  - Agents' Behavior: because the market cannot be relied upon across time, agents retain food self-sufficiency as a non-negotiable baseline, supplementing it with household crafts and occasional trade. This is not backwardness but rational diversification under endemic uncertainty.
  - Capital Behavior: in a volatile market, the factory system faces a structural barrier that makes large-scale production economically irrational: a factory requires high overhead—rigid commitments to wages, rent, equipment, and debt service. In an environment of recurrent shocks, these fixed costs become a structural liability: repeated demand slumps, typically triggered when spikes in grain prices drain purchasing power for industrial goods, progressively exhaust cash reserves and eventually lead to insolvency. In contrast, the small-scale household survives through its flexibility; with no rigid overhead, it can retreat to self-sufficiency during downturns that would destroy a larger enterprise burdened by fixed costs. Because industrial production is vulnerable to price noise and unpredictable shocks, its risk-adjusted returns are too low to justify the commitment of capital. Rational capital thus avoids the industrial sector, flowing instead into land, short-term lending, or hoarding, which effectively blocks industrialization.

## 3. Institutions: Context-Specific Adaptations

Based on the micro-mechanism above, we must re-evaluate institutional differences across regions. The key insight is that institutions are rational adaptations to their respective environments.

- **Rethinking Sharecropping:**
  - The conventional view: sharecropping has long been criticized for its weak incentive properties, as the tenant receives only a fraction of the marginal product, leading to suboptimal effort. Later literature reframes it as a trade-off between incentives and risk-sharing, taking the level of output risk as given. Yet this literature treats the level of risk as an exogenous parameter, never asking where it comes from or how its magnitude transforms the very nature of the problem.
  - Our refinement: in highly volatile environments where output volatility is

driven by severe climatic shocks, the problem is not merely balancing incentives against risk-sharing. Here the primary institutional challenge is the asymmetry of risk: while the upper bound of harvests in good years is relatively stable, the lower bound during disasters is unpredictable and often drops below subsistence levels. Under such conditions, rigid fixed-rent contracts become effectively unenforceable. Attempting to enforce a fixed payment during a major harvest failure does not lead to wealth accumulation, but to social friction, disputes, and potential violence. Because the risk of tenant insolvency is recurrent, landlords and tenants adopt sharecropping to provide the contractual flexibility necessary to maintain social order and long-term land productivity.

  - The Western counterpart: England's widespread use of rigid fixed-rent contracts has long been celebrated as institutional superiority, but is better understood as a reflection of its geographic stability. In a stable environment, the risk of a tenant being unable to pay is minimal, allowing the law to be enforced as written with low enforcement costs.

- **Kinship Networks and Social Structure:**
  - The Rationality of Kinship Systems: in a highly volatile environment, where state-provided relief is insufficient and social order is unstable, the kinship system serves as the primary framework for risk-pooling and survival. Even in regions like the Yangtze Delta, where individual households may appear to function as nuclear families, agents still relied on broader kinship networks for mutual aid during shocks and collective security during crises. These institutions may appear to be cultural preferences, but at a deeper level they are rational adaptations to geographic volatility.
  - The Western Divergence: In a stable environment, the natural predictability of output reduces the necessity of kinship-based protection. This enables a more atomized social structure, with the individual or nuclear family becoming the primary economic unit. This atomization is a consequence of geographic stability, and it significantly lowers the barriers to labor mobility. It enables individuals to seek specialized industrial work, free from the bonds of kinship.
- **Governance Paradox:**
  - The structural mismatch: a highly volatile environment generates a high demand for state intervention (disaster prevention, domestic order, and external defense) while simultaneously undermining the fiscal capacity to fund it as the tax base becomes both volatile and difficult to assess. Faced with this paradox, states typically follow divergent paths: evolving into coercive fiscal states to survive, or suffering collapse due to fiscal failure.
  - The Western exception: England enjoyed a double blessing: geographic stability minimized the need for disaster prevention and lowered social friction, while the island geography reduced defense costs. This minimized the transaction costs of social coordination, making a limited government viable. In highly volatile environments, such a model would have been untenable.

## 4. Division of Labor: The Intertemporal Constraint

These foundations allow us to refine Adam Smith's fundamental proposition: "The division of labor is limited by the extent of the market." Traditionally, the extent of the market is analyzed through geographic reach or population size. We add a third dimension: its stability over time.

Even if a market is spatially vast and integrated (like traditional China), it can be fragmented in the temporal dimension. Frequent natural shocks cause extreme price volatility, which periodically disrupts market transactions and breaks the continuity of the market. For the individual, the division of labor means specialization. **Surplus enables specialization; only stable surplus enables full specialization.** Where surplus is inherently unstable, rational agents do not fully specialize. They instead choose within-household self-sufficiency as a form of self-insurance. This constrains specialization, trapping the economy in a stagnant pattern of self-sufficiency.

## 5. The Material Threshold of Industrialization

In summary, industrialization is not just a matter of capital accumulation, technological innovation, or the design of incentive mechanisms; it is, at its core, a process of de-risking. Only when output volatility is tamed can a well-functioning market take root, allowing capital to accumulate, technology to advance, and incentives to function as designed.

- **The Western path (providence):** the West benefited from a providential environment that provided both favorable and predictable grain output. This was the binding advantage: while other factors (transport, energy, geopolitical security) also mattered, it was agricultural stability that allowed well-functioning markets to emerge and industrialization to unfold spontaneously.
- **The Chinese path (engineering):** faced with inherently high grain output volatility, a purely market-based system would struggle to sustain the intertemporal stability required for industrialization. This demanded a proactive state to undertake systemic engineering to mitigate the effects of geographic volatility. Through these material investments, the state engineered the conditions for a well-functioning market and a subsequent industrial takeoff.

# Historical Analysis

## I. Origins: Law and Engineering

At the formative stage of their respective civilizations, two legendary events set the course for the divergent trajectories of governance in the West and China.

The first centers on Mount Sinai, where Moses ascended alone to receive divine revelation. He brought down not food, which was already a providential gift, but the Ten Commandments. In the wilderness, Moses established order through **"The Law"**: a covenant between man and God, and by extension, between people. This represents the preservation-oriented governance of the West, where order is seen as a divine inheritance. Centered on the preservation of eternal rules, it presupposed a well-provisioned sanctuary—an 'oasis' where the primary governance task is settling disputes about rights. In the modern West, this has evolved into a deep reliance on the rule of law.

The second unfolds at Mount Tu. Here, Yu the Great, the legendary sage-king and founder of China's first dynasty, gathered the tribal leaders, but only after thirteen years of wading through mud to tame catastrophic floods. Faced with nature's chaos, he did not wait for divine intervention. Instead, he mobilized the people to dredge rivers and level hills, physically reshaping the landscape. Yu the Great established order through **"Engineering"**: transforming nature to define the realm. This represents the transformation-oriented governance of China, where mastery over environmental volatility provides the essential platform upon which social order and rules of governance can subsequently be built. In modern China, this logic is manifested in an engineering-led state.

For centuries, the rise of the West persuaded the world that rights-focused governance was the only path. Today, China's successful Big Push reasserts an alternative logic of engineering-led governance.

## II. Geography and the Survival Challenge

At its core, human history is the story of how people respond to nature's challenge of survival and development. Different geographic environments set different rules for survival, shaping distinct civilizations.

Chinese civilization was born from this very challenge. Its land is far from barren; instead, it possesses vast, fertile plains that have supported the world's largest population. For most of history, this natural wealth allowed China to maintain the world's largest economy.

Yet, this abundance came with a dangerous catch: high geographic volatility. Driven by the continental East Asian monsoon, the unpredictable timing and volume of water translate into

significant fluctuations in grain output. Rarely has a civilization been so defined by its struggle against nature, yet endured for millennia. From the Dujiangyan irrigation system to the Grand Canal, China's engineering miracles were essential material investments to survive in a highly volatile environment.

This constant struggle forged a tradition of practicality over dogma. In the face of ruthless natural laws, empty rhetoric offers no protection. Survival required massive organization and physical engineering. In China, survival was never a providential gift; it was an outcome earned through persistent effort in the face of recurring crises.

Nevertheless, while this immense survival pressure stimulated China's potential to transform nature, it also built an invisible wall. The constant struggle just to survive locked the society into a stagnation trap, blocking the spontaneous path to industrialization.

When the Industrial Revolution arrived, the glory of this agrarian civilization suddenly became a liability. An agricultural nation at the mercy of the monsoon could not withstand an industrial power armed with machines. This, fundamentally, is the essence of China's "century of humiliation"—not a cultural failure, but the collapse of a pre-industrial society facing industrialized military powers.

To truly understand why industrialization failed to emerge spontaneously in China, we must first dispel a common misconception. Historians often blame this stagnation on overpopulation or despotic institutions. But these are symptoms, not causes.

Why was there overpopulation? Because in a highly volatile environment, a large family was the primary mechanism of self-insurance against crisis. Why was the state intrusive? Because in a highly volatile economy, the immense difficulty of reliably monitoring and collecting revenue forced the government to be coercive in securing its fiscal base.

Volatility is the root; population and institutions are merely adaptations required to survive it. We can only understand the true threshold of industrial civilization by grasping this reality.

## III. Foundations of a Well-Functioning Market

Industrialization does not happen in a vacuum; it requires specific social and economic preconditions. These conditions are not providential gifts, except for a few lucky nations. The most critical prerequisite is the formation of a well-functioning market. In economic theory, a well-functioning market is defined by its ability to coordinate resources through reliable price signals and predictable rules. While mainstream economic models routinely assume the clarity of price signals (and even where the literature examines signal distortion, it typically attributes the problem to information asymmetry between agents), the role of the physical environment itself as a structural source of price instability has not been adequately incorporated into the theoretical frameworks that underpin development economics. This is precisely where theory parts from reality.

A well-functioning market does not appear simply by adopting a constitution or hanging up a “free market” sign. Its prerequisite is stable access to essential material supplies: the clarity of price signals, and even the predictability of rules, ultimately rest upon material fundamentals. In pre-modern societies, food was the essential supply. In the modern era, energy has joined it.

Across all eras, however, the supply of food remains the very foundation of economic life. While the volume of grain output defines the scale of the market, the volatility of grain output determines the degree of systemic stability. As Emperor Qianlong observed, grain prices set the anchor for all other prices. When this anchor holds firm, prices transmit clear value signals rather than survival panic, and contract enforcement remains straightforward. This provides the stable expectations required for specialization and long-term commitments. In contrast, in a highly volatile environment, the entire price system becomes unreliable, and contract enforcement grows costly as defaults become endemic; stable expectations are destroyed, rendering specialization and long-term commitments unviable. In this sense, the "invisible hand" of the market is only effective when it rests upon the solid ground of material predictability. Without this stability, all rational individual decisions point toward hedging for survival, effectively blocking industrialization.

## IV. Providential Gift: The Rise of the West

The rise of the West was multidimensional, but fundamentally, it rested on the overwhelming material power generated by the Industrial Revolution. England, which nurtured this revolution, was a special case where a well-functioning market formed and industrialization unfolded spontaneously. However, this was not the inevitable result of some abstract culture or institution, but the dividend of a geographic jackpot: a rare case where agricultural stability, energy abundance, easy transport, and strategic security converged. While other regions had their own resources, England enjoyed the providential luxury of predictability—a decisive advantage that required no prior physical engineering.

### 1. The Geographic Jackpot

England possesses a mild and moist maritime climate. Influenced by the North Atlantic Drift and westerlies, rainfall is reliable year after year and stays remarkably even throughout the seasons. This favorable stability is exceptional. Just across the sea, Ireland and Western Scotland suffer from excessive rainfall that makes wheat farming precarious; to the south, Mediterranean France experiences higher volatility; to the east, the North German Plain contends with harsher winters. But Southeastern England, along with the Low Countries and northern France, sits in a perfect “Goldilocks zone.” In sharp contrast to the volatile monsoon climate of China, English farmers were spared the violent cycle of feast or famine. While mean yields might not have matched the intensive rice paddies of East Asia, the volatility, the binding constraint, was exceptionally low.

It was this natural stability in grain production that allowed England to embark on a path different from that of the vast continental and monsoon regions of Eurasia. Here, governance

could focus primarily on delineating boundaries and writing leases. Grain harvest was reliable; effort translated directly into reward. Unlike the Yangtze Delta, where stable yields required enormous investment in irrigation, drainage, and flood control, England's agricultural stability demanded minimal hydraulic engineering—a natural gift that delivered a better result without the crushing cost.

## 2. Fixed-Rent Tenancy and Predictability

To understand the rise of the West, we must look closely at the subtle but decisive difference in the baseline institution: land tenure.[2] While economic historians often note that fixed rent (paying a set amount regardless of harvest) is the hallmark of modern commercial farming, this institution was not unique to England.

In advanced agrarian regions like the Yangtze Delta in China, fixed rent (known as *iron-rent*) was a common legal practice, and perpetual tenancy (known as *yongdian*) also existed in certain localities. However, the same institutional form produced completely different results on the two ends of Eurasia. The difference lay not in the clauses of the law, but in the physical volatility of the environment in which they were enforced.

**Risk-sharing in China: A Flexible Survival Mechanism**

Under the continental East Asian monsoon, where floods and droughts were frequent, a rigid fixed rent was effectively unenforceable. If strictly implemented, a bad harvest would lead to mass tenant default and violent uprisings. Therefore, Chinese society evolved rent abatement practices, sometimes tacit custom, sometimes mandated by official policy, for times of famine. This was not a Chinese peculiarity but a practice widely attested across pre-modern agrarian legal traditions: Roman jurists had long codified the principle of *remissio mercedis,* rent abatement for harvest failure caused by forces beyond the tenant's control, on the ground that no contract can justly bind a party against the force of nature. Whenever disaster struck, tenants had the right to demand a rent reduction. Consequently, the fixed rent was nominal. In reality, it constantly changed with the fluctuations of nature, effectively reverting to a form of sharecropping, where risks were shared. Furthermore, perpetual tenancy (*yongdian*) was a legal defense against an environment that constantly threatened to dispossess the tenant, yet even this legal shield crumbled when the survival crisis forced tenants to sell their rights just to buy food. Legal stability collapsed under economic volatility. The contract was not an anchor for capital accumulation, but a flexible buffer for tenant survival.

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[2] We call land tenure the "baseline institution" because it occupies a unique position in the institutional hierarchy: it is the interface through which the physical constraints of nature are translated into social arrangements. In agrarian societies, land tenure governed the distribution of the primary source of wealth, and the arrangements it fostered shaped the evolution of other institutions from contract law to state finance. Land tenure is indeed the oldest institution to have accompanied civilization since its origin, but its primacy is not merely historical. Agriculture is not merely the starting point of civilization but its permanent foundation. Even in industrial societies, the entire edifice of specialized production rests on a premise that must be continuously satisfied: participants in the division of labor do not need to grow their own food. When that premise fails, all higher-order institutions collapse. Land tenure is "baseline" because it governs this foundation, and will remain so as long as human survival depends on a stable food supply.

**Rigidity in the West: The Engine of Development**

England, however, enjoyed the geographic jackpot of stability. Here, the English fixed rent was a different species: it was genuinely rigid. Because extreme harvest failures were rare, famine abatement clauses were largely absent from the system. Contracts were enforced as written. Where abatement occurred, it was either stipulated in individual leases or granted voluntarily by landlords. It was not a systemic legal principle, but an occasional private arrangement that the low frequency of harvest failure made rarely necessary. This seeming rigidity was actually a blessing. It meant that the tenant farmer, after paying the fixed rent, became the residual claimant of all surplus. And tenure security followed the same logic. In England, the standard arrangement was not perpetual tenure but fixed-term leases—typically of 7 to 21 years. No legal guarantee of permanence was needed: low output volatility ensured that tenants reliably paid their rent and landlords reliably renewed the lease. Tenure security was a dividend of geographic stability, not a legal invention. Predictability is the foundation of long-term investment and capital accumulation: Knowing that "nature will not steal my surplus," English farmers dared to invest heavily in land improvement and machinery while retaining the gains year after year. Thus, capital accumulated. The English fixed rent was not a superior invention of human wisdom, but a privilege of predictability, a luxury that only a providential oasis could afford.

**The Exception that Proves the Rule: Dujiangyan and the Po Valley**

The case of the Chengdu Plain in China, governed by the ancient Dujiangyan irrigation system, provides a crucial exception that confirms this risk-based logic. In this local oasis where engineering mastered the natural volatility, the fixed rent (*iron-rent*) indeed became as genuinely rigid as its English counterpart. Because the risk of harvest failure was virtually eliminated, land contracts in this region achieved genuine rigidity: famine abatement clauses were either absent or, where present, rarely invoked in practice. It fostered a wealthy class of managerial farmers, proving that Chinese agents were fully capable of developing rigid contractual institutions once the physical environment allowed for it. However, this local success failed to ignite a national industrial revolution. Such stable zones were extraordinarily scarce and small. They simply could not overcome the overwhelming burden of continental volatility. A comparable process occurred in the Po Valley of Northern Italy. By the late 19th century, its core engineered stability zone spanned approximately 10,000 square kilometers. This scale fostered similar fixed-rent contracts and managerial farming modes across a vast contiguous belt. At the turn of the 20th century, this zone was large enough and sufficiently connected to Mediterranean trade to accumulate the capital and demand necessary for Italy's northern industrial takeoff. In contrast, the Dujiangyan core remained a geographically isolated sanctuary of about 2,000 square kilometers. Hemmed in by the mountains of the Sichuan Basin, it remained a "brilliant but isolated dot" on a vast, volatile continent.

## 3. Rise of Tenant Farming

Built upon this foundation of rigid, predictable fixed rents, England developed the tenant-farming system to its fullest extent. As an operator, the tenant farmer rented land at a fixed rent, hired laborers at fixed wages, and retained the residual surplus. This business model was the precursor

to the modern enterprise. It follows the same logic: paying fixed interest to banks, fixed wages to workers, and claiming residual profit.

However, this structure implies high operating leverage. For a business burdened by rigid fixed costs (fixed rent and wages), revenue stability is a matter of life and death. In highly volatile environments, the challenge is not just an occasional bad harvest, but structural volatility. The incessant swings between feast and famine make future cash flows impossible to predict. Under such uncertainty, adopting a high-fixed-cost structure is economically unviable; recurrent lean years would eventually drain liquidity and bankrupt the operator.

This explains why large-scale commercial farms failed to emerge in traditional China despite high land productivity. In England, however, nature acted as the ultimate insurer, smoothing revenue streams and allowing this high-leverage model to thrive. While family farms persisted and scale had its limits, the dominant trend was the consolidation of land into efficient, capital-rich units. Thus, tenant farming gradually grew and prospered, turning agriculture from a survival activity into a capital-accumulating industry.

While tenant farming was England's distinctive path, the same underlying logic operated across the stable zone in varied forms. In the Paris Basin, large fixed-rent farms closely mirrored the English model. In the Low Countries, monetized fixed-rent tenancy predominated, embedded in a highly commercialized economy. In the American North (New England and the Mid-Atlantic), independent freeholders were the norm, though tenancy was not absent. The specific agrarian forms varied, but the common thread was the viability of fixed-cost arrangements in a stable environment: obligations could be met reliably, and a substantial share of the surplus could be retained and reinvested.

## 4. The Foundation of Social Coordination

The natural stability of grain output brought profound social consequences. Adam Smith, drawing on the English experience, proposed three elements for wealth growth: "Peace, easy taxes, and a tolerable administration of justice." For centuries, this has been viewed as the universal institutional recipe for prosperity. But in reality, these are simply the institutional dividends of the geographic lottery:

- **Peace:** Not just the absence of war, but low internal social friction. When the future is predictable, the desperation for survival vanishes. Society shifts from a zero-sum game of scrambling for limited resources to a positive-sum game of continuous accumulation. As Mencius observed, "Those with a secure livelihood have a steady mind." Peace is not a moral achievement, but a dividend of physical stability.
  In England, the Low Countries, northern France, and the American North, secure livelihoods lowered social friction and freed individuals from reliance on lineage-based security. They lived in nuclear families, standing in marked contrast to the kin-based extended families of highly volatile regions. This social structure allowed people to leave the land for industry, leading to lower birth rates and a more balanced population

structure. This, in turn, promoted agricultural progress and paved the way for industrialization.

- **Easy Taxes:** In highly volatile regions, tax collection becomes a constant struggle. With a tax base both volatile and difficult to assess, the state naturally suspects taxpayers of concealing wealth, while taxpayers naturally fear the state taking too much. This mutual distrust forces the state to intrude constantly upon private property to verify and seize resources, effectively blurring the line between taxation and confiscation.

  In England, however, under peaceful conditions, the government did not need heavy taxes. Internally, low social friction reduced the need for suppression; externally, island geography minimized the need for a standing army. The predictability of harvest made the tax base transparent. The government could collect a fixed revenue without intrusive monitoring, leaving property boundaries undisturbed. Property rights protection was not a result of state benevolence, but a consequence of low transaction costs in taxation.
- **Tolerable Administration of Justice:** This point is the most misunderstood. The fundamental reason England could practice a loose and piecemeal common law system was that social friction was naturally low.

  In highly volatile regions, frequent disasters turn every contract into a dispute (e.g., tenants demanding rent abatement), generating a massive demand for judicial intervention and disaster relief. Beyond justice and relief, volatility also demands heavy investment in disaster-prevention infrastructure (dikes, granaries) and social order maintenance during crises. This creates a "Governance Paradox": it generates a massive demand for public services while also weakening the fiscal capacity to fund them. With a tax base both volatile and difficult to assess, the state lacks the fiscal resources to maintain a professional bureaucracy, forcing it to rely on low-cost local proxies such as lineages or gentry to extend its reach. While this maintains basic order, it is a far cry from the efficient, uniform, direct administration imagined by modern observers.

  England, conversely, enjoyed a double blessing: low social friction reduced the cost of governance, while stable revenues ensured the capacity of governance. Justice could remain merely "tolerable", loosely structured, reactive, and locally administered, because the system was rarely pushed to the breaking point. It possessed the fortune of not needing a heavy hand, rather than the wisdom of designing a limited government.

England enjoyed these conditions in their purest form. Across the Channel, the Low Countries and northern France shared much of the same stability, though with somewhat higher fiscal burdens imposed by continental defense. The foundation of social coordination was not uniquely English, but English conditions were uniquely favorable.

The economic essence of the English success was the presence of a stable "grain anchor." In economic terms, this material predictability structurally reduced the transaction costs of social coordination. A well-functioning market is, at its core, an information system. When the price of food is steady, the signals across the entire price system are clear and reliable. This clarity dramatically lowers the costs of information gathering, contract negotiation, and contract enforcement across all sectors.

This foundational anchoring is the invisible prerequisite for the rule of law: material predictability must precede legal certainty. The conventional institutional perspective treats "better laws" as an exogenous gift of wisdom and an independent cause of growth, but in reality, these "better laws" are the institutional dividends of a stable environment, and gain efficacy because the transaction costs of enforcing them are low enough to make enforcement viable. In a highly volatile environment where food prices are chronically unstable, the resulting chaos renders even the most sophisticated legal codes unenforceable.[3]

## 5. Spontaneous Industrialization

Before the Industrial Revolution ignited, the Agricultural Revolution had already prepared the essential fuel: a massive food surplus. In a stable environment, farmers of all kinds—freeholders, tenant farmers, large and small—had incentives to invest in land improvement. The spread of the four-course rotation (wheat, turnip, barley, clover), which eliminated fallow periods and boosted yields, was one prominent result. However, this technical leap was not merely a farming improvement; it was a dividend of stability. The system required dedicating significant land to fodder crops (turnips, clover) for soil regeneration and livestock. In highly volatile monsoon regions, where every inch of land was needed to grow grain for human survival, dedicating land to fodder crops instead of grain was a luxury few could afford. Even where rotations existed (e.g., rice followed by green manure crops), they were aimed at maximizing caloric density, not growing different crops for the market. This specific constraint reflected a broader reality: the constant threat of famine forced a focus on immediate caloric extraction, blocking not just commercial crop rotation, but any form of market-oriented farming that required long-term stability.

While agrarian stability was the foundation, the geographic lottery endowed England with three additional natural advantages:

- **Geopolitical Security:** The English Channel served as a natural moat, insulating the island from the continental wars that periodically destroyed wealth in Europe.
- **Low Transport Costs:** No place in England is more than 120 km from the sea. A dense network of rivers and canals connected industrial centers (like Manchester) with ports at extremely low cost, unifying the domestic market.

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[3] The same material logic determines the type of Leviathan a society can sustain. Acemoglu and Robinson's *The Narrow Corridor* classifies Leviathans as Shackled, Despotic, or Absent, defined by the balance of power between state and society. This typology accurately describes the phenotype of governance, but leaves the genotype unidentified: it does not explain why some societies achieve balance while others do not. Within the mean-variance framework, the answer is straightforward. The Shackled Leviathan is the institutional expression of a low-variance environment: stable output minimizes the demand for state intervention, so the state can govern without expanding, and society can restrain it without confrontation. The Despotic Leviathan reflects a high-mean, high-variance environment: volatility simultaneously inflates the demand for state action (disaster prevention, famine relief, domestic order, external defense) and erodes the fiscal base, forcing the state into coercive extraction. The Absent (or Paper) Leviathan marks the low-mean, high-variance quadrant: the state cannot even sustain itself, let alone provide public goods. The "narrow corridor," therefore, is not a political achievement that any society can enter through wise institutional design; it is the governance space that low variance opens up. England did not "shackle" its Leviathan through parliamentary ingenuity; it possessed the fortune of not needing a strong Leviathan in the first place.

- **Energy Abundance:** Vast, shallow coal deposits were located conveniently near the coast (e.g., Newcastle), providing cheap, accessible fuel for the coming Steam Age.

At this point, a unique historical convergence occurred. A stable and abundant food supply secured the workforce; abundant coal provided cheap power; water transport ensured low-cost circulation; and island geography protected the accumulation of wealth. When these factors came together, market uncertainty and production costs reached a historical low.

On this material foundation, a well-functioning market formed naturally. Capital was no longer confined to landholding or commerce. With the assurance of a stable market, wealth owners dared to tie up their liquid resources in fixed assets—factories and machines. Industrialization was not a top-down design, but the natural emergence of a system where the material threshold had already been crossed, allowing stable markets and long-term industrial investment to flourish.

England was the first to cross this threshold, but the same convergence of stable agriculture, cheap transport, and accessible energy replicated itself across northwestern Europe and the American North, making the rise of the West a broad-based phenomenon rather than an isolated English miracle.

## 6. The Supersized England: The American North

The American North was another winner of the geographic lottery, an enhanced version of the English pattern.

New England and the Mid-Atlantic states (like Pennsylvania and New York) share a similar temperate, humid climate. Although situated in a continental zone, the region enjoys a rare degree of climatic stability. Thanks to the moderating influence of the Atlantic Ocean (and, near the lakeshores, the Great Lakes), rainfall is relatively reliable year after year. While seasonal amounts vary, the timing is steady and predictable, free from the wild fluctuations of the monsoon. Thus, the American North was blessed with agricultural stability. In states like Pennsylvania, known as the breadbasket of early America, abundant yields and predictable prices were the norm. This predictability was the foundation of its social order. Socially, while the prevalence of independent freeholders originated from colonial land policies, their ability to stay independent was a dividend of agricultural stability.

Unlike in highly volatile regions where frequent crop failures triggered a cycle of debt and insolvency, the agricultural stability meant that a single family could be a self-sustaining economic unit without needing to surrender its autonomy for protection. This prevented the slide into debt bondage—the recurring cycle of debt and dispossession that once weakened the Roman Republic and has plagued agrarian societies throughout history. Instead, this stability cultivated a resilient class of independent farmers who provided the stable social foundation and consumer market for the coming industrial age. This stands in stark contrast to the American South, where greater climatic variability compounded by reliance on volatile cash crops created

chronic fragility. Rather than a resilient yeoman class, this environment produced a marginalized underclass of "poor whites"—smallholders on marginal lands who, lacking the reserves to buffer against crop failures and volatile market prices, were unable to sustain their independence. After the Civil War, this highly volatile environment led to the widespread adoption of sharecropping, trapping both former slaves and poor whites in systemic poverty for nearly a century.

The natural setup for transport and power was equally lucky. Excellent natural harbors like Boston, New York, and Philadelphia dotted the coast. Inland rivers like the Hudson were navigable deep into the interior. The opening of the Erie Canal in 1825 connected the Atlantic to the Great Lakes, slashing transport costs. While New England lacked coal, it had powerful rivers for hydropower; nearby Pennsylvania had vast coal reserves to fuel the later Steam Age.

Stable and abundant food supply, abundant energy, cheap transport, and isolation from European conflicts—the American North perfectly replicated England's material foundation. With a vast hinterland, it became a supersized England. Here, a well-functioning market formed spontaneously, and industrialization unfolded as a natural progression.

## V. Colonial Plunder: A Boost, Not the Engine

While the conventional institutional perspective seeks the cause of Western success in human wisdom, an alternative and widely influential narrative seeks it in human sin, arguing that industrialization was fueled primarily by colonial plunder.

History proves that colonial plunder alone cannot make a modern nation strong; Spain is the most telling example. Comparing the failure of Spain with the success of England is not about excusing the sins of the past. It is a lesson in how nations actually grow: external wealth only works if a nation already has a stable foundation at home.

As the first empire on which the sun never sets, Spain plundered an astonishing amount of gold and silver from the Americas following Columbus's voyage of 1492. If colonial plunder were the engine of industrialization, Spain should have become the world's ultimate industrial powerhouse. Instead, it became merely a sieve for gold.

Spain's heartland, the arid central plateau and Mediterranean coast, suffered from high grain volatility. Without a predictable food supply, it could not support the long-horizon commitments required for a robust industrial base. Therefore, the massive wealth from the Americas was not invested in domestic production. Instead, it was dissipated on imports, conspicuous consumption, and costly wars. Gold flowed through Spain like water, enriching the Netherlands, Britain, and France—countries that possessed the internal stability Spain lacked. Sitting on mountains of gold, Spain failed to industrialize in step with its rivals and eventually declined into a second-rate power. Tellingly, the one Spanish region that did industrialize spontaneously, the Basque Country, is precisely the one that combines a maritime climate with low grain volatility, mirroring the English pattern. Yet like the Dujiangyan oasis, it was too small to overcome the volatility that held the rest of Spain back.

In contrast, England developed a prosperous domestic economy (textiles, shipbuilding) based on agricultural stability before it ever became a global plunderer. When England later turned to empire-building, every penny snatched from abroad could be invested into a thriving local industrial cycle: building factories, upgrading technology, and expanding trade. England became the global industrial center not just because it had the claws to snatch wealth, but because it had the “stomach” to digest and multiply it.

Colonial plunder was undoubtedly a huge boost for the West, but it was not the engine. The true engine was a convergence of geographic advantages: stable and productive agriculture, convenient water transport, and abundant energy. The regions surrounding the English Channel (England, Netherlands, Belgium, northern France) and the American North shared this winning combination. With this engine, the West acquired the capacity to plunder the globe.

To attribute the rise of the West merely to theft is morally satisfying but logically flawed. This focus on external causes hides the true internal threshold of industrial civilization. The Global South, lacking this jackpot, cannot simply replicate the Western path. China faced the same constraint—and had to forge a different way forward.

## VI. Traditional China: Trap of Stagnation

What Chinese civilization faced was not just natural abundance, but a brutal challenge imposed by the continental East Asian monsoon climate. Here, boundaries and contracts meant nothing against the recurring shocks of floods and droughts.

### 1. The Fragility of Specialization

An ancient song traditionally attributed to Emperor Shun captures the essence of the monsoon:

> The fragrance of the South Wind eases my people’s anger.
> The timing of the South Wind increases my people’s wealth.

This song reveals the two sides of the monsoon. When the "south wind" (monsoon) arrives on time, it brings rain and heat, creating immense wealth. But the monsoon is fickle. Once it is untimely, abundance turns into disaster. For millennia, this uncertainty of living at the mercy of the monsoon has been a constant threat to survival. But it was more than that: it was also a barrier to specialization.

Take the salt industry as an example. If salt workers focused only on making salt and traded with farmers for food, everyone would benefit: specialization raises efficiency, and exchange allows both sides to gain. But in reality, this specialization was fragile.

For the salt worker, the monsoon created a constant threat: In lean years (drought or flood), grain harvests failed and prices skyrocketed. Those who had to buy grain spent most of their earnings just to survive. Farmers fared no better—a failed harvest meant no surplus to sell and no income

to spare. Either way, demand for salt collapsed. The salt worker was trapped: grain was unaffordable, and salt found no buyers. To avoid starvation, he dared not specialize in salt; he had to grow some food himself as a lifeline.

Historically, the government tried to force specialization by mandating hereditary occupations (e.g., salt workers) and establishing large-scale state-controlled workshops. But this forced specialization was equally fragile. It relied on the state's ability to guarantee food rations. When the state's fiscal capacity was strained by famine or war, the lifeline of grain became unreliable. These workers had to make do with inadequate relief or simply fled. Whether forced by the state or attempted by the market, specialization without a guaranteed food supply was a castle built on sand.

The same logic applied to the farmer. In bountiful years, grain prices crashed; in lean years, there was no surplus to sell. So he remained trapped in poverty. He couldn't focus solely on farming; he had to weave cloth or do odd jobs to make ends meet.

This logic extended to all livelihoods. Workers dared not specialize; farmers dared not specialize. Fundamentally, specialization requires a stable exchange ratio; high volatility destroys it.

## 2. Why No Factories?

People often blame stagnation on overpopulation. But this is only a symptom. If this massive population could be absorbed by factories, that would be a dividend, not a curse. Why factories failed to absorb this workforce needs a more fundamental explanation.

Take the traditional cotton textile industry as an example. Before the Opium War, China produced an estimated hundreds of millions of bolts of cloth annually. This seems like a huge industry. But this was not the result of mass production. It was done by countless family workshops and consumed locally.

Why did large textile factories fail to emerge in this massive sector? The fundamental reason was high geographic volatility, which killed the possibility of expansion in scale and capital accumulation.

**First,** on the demand side, high market volatility made large-scale operations economically unviable. In an economy of "feast or famine," purchasing power fluctuated in lockstep with the harvest.

- In bountiful years, when food was cheap, families had money left over to buy cloth.
- In lean years, people couldn't even afford food. Demand for cloth would crash.

These recurring shocks meant that any factory owner who invested in machines and hired workers faced a severe threat: frequent demand slumps, triggered when spikes in grain prices

drained purchasing power, made it impossible to sustain cash flow. He would eventually be buried by unsold inventory and fixed costs.

Second, on the supply side, capital could not be accumulated from the bottom up. Large-scale production requires sustained reinvestment. But in this environment, small producers spent most of their earnings buying expensive grain in lean years just to survive. The surplus created in bountiful years was eaten up by the deficit of lean years, preventing the accumulation of capital to build factories.

Finally, even those with capital (such as landlords or merchants) dared not invest in factories. Given the high uncertainty, they preferred land, short-term lending, or hoarding, which offered quicker returns or tangible security. Industrial capital was effectively blocked from forming.

The result was a structural impasse imposed by high volatility. As Adam Smith said, "the division of labor is limited by the extent of the market." In traditional China, the binding constraint on market extent was not geographic reach or population size, but high volatility. In such a market, the factory system was economically irrational for all parties: workers could not afford to leave the land, and capital avoided being tied up in factories. This explains why the workforce remained stuck in households and why the Industrious Revolution (the intensification of household labor) in China never led to an Industrial Revolution.

## 3. The Self-Reinforcement of Involution

High geographic volatility, through its effect on grain output, also pushed population growth to the limit through two dynamics, intensifying population pressure on land.

First: Tied to the Land. Because of frequent floods and droughts, farmers dared not leave the land. Otherwise, they might not be able to afford food in lean years. Therefore, no matter how much the population grew, everyone had to cling tightly to the land as a lifeline.

Second: The family safety net. Since life was hard and crowded, why have more children? Because in a highly volatile environment, the family was the primary safety net. The belief that "more children, more blessings" was not a superstition, but a rational survival strategy:

- **Labor and mutual aid:** More hands meant not only more help in the fields, but also a lifeline during disasters. When crops failed, the extended family pooled their food and labor to get everyone through the crisis.
- **Collective defense:** A large extended family was not just for mutual aid, but also a powerful deterrent. In a world where law and order were weakened by the chaos of nature, such numbers discouraged bandits, thieves, or even greedy neighbors from preying on your household. In a highly volatile environment, there was truly safety in numbers.

These two dynamics created a trap far more complex than simple poverty: “Involution.”
Intensive labor input with diminishing marginal returns: as more hands worked the land, output

per person declined. Farmers farmed and wove at the same time. The whole family worked long hours with high intensity, yet they remained extremely poor. This was not because they were lazy or stupid, but because specialization was economically unviable in such a highly volatile environment. Worse still, involution left the system dangerously fragile: a dense population surviving on minimal surplus, with no buffer against consecutive bad harvests. Individual strategies for risk-minimization aggregated into the systemic fragility. As R.H. Tawney observed of rural China: "There are districts in which the position of the rural population is that of a man standing permanently up to the neck in water, so that even a ripple is sufficient to drown him."

While this tendency was widespread across highly volatile environments, it found its most extreme expression in rice-growing regions like the Yangtze Delta. Unlike wheat, wet rice cultivation acts as a "labor sponge"—it allows for an exceptionally high degree of labor intensification (e.g., hydraulic management, heavy fertilization) to squeeze out marginal gains.

This immense labor input did, in fact, mitigate the physical volatility of yields and allowed for significant capital accumulation. The region became wealthy, and a commercial class emerged. But this wealth did not arise from large-scale farming or factory production. It accumulated through traditional channels: landlords collecting rent, merchants profiting from long-distance trade in silk and cotton cloth, and moneylenders earning from short-term credit.

Given that intensive labor had already mitigated yield volatility in the Yangtze Delta, and that wealth had accumulated, why did this capital never flow into factories? The answer lies in involution, the survival strategy itself, which came at a crushing economic cost: the devaluation of labor. Chronic labor surplus, aggravated by agricultural seasonality, pushed the price of labor to the subsistence floor. This cheap labor neutralized the power of the accumulated capital to trigger industrialization:

- **The Fragility of Demand (Demand Side):** Low labor value meant that despite the region's aggregate wealth, the vast majority of households lived with minimal financial buffer. They remained extremely sensitive to grain price volatility. A slight rise in food prices would wipe out their purchasing power for industrial goods, making the mass market too unstable for factory investments.
- **The Capital Diversion (Supply Side):** As we will explore next, the abundance of cheap labor destroyed the economic rationale for machinery investment. Even though capital was available, it did not flow into machinery—because machines were expensive while men were cheap.

Thus, the economy achieved "Growth without Development": capital accumulated, but the very conditions that enabled it blocked its industrial outlet.

## 4. Rational Regression

It is generally recognized that in traditional China, the smallholder economy blocked industrial progress. But there were exceptions—moments when advanced machinery did appear. These

leaps were typically driven by the state or local elites, who possessed the capacity to concentrate capital and mobilize resources for high-level engineering. As one of these “technological peaks,” the water-powered Grand Spinning Wheel, documented during the Yuan Dynasty, represented the technological frontier of the agrarian era. Used for spinning hemp, it had 32 spindles and was driven by a massive water wheel. Its complexity was notable.

But the seeds of stagnation were sown in this very prosperity. By the Ming Dynasty, two things changed: cotton replaced hemp, and driven by the involution trap, population pressure surged to a saturation point. As we explained, the involution trap forced people to stay on the land and reproduce, driving the market price of labor to the subsistence floor. Since human labor was cheaper than maintaining a complex machine, labor-saving technology became unviable. Although the Grand Spinning Wheel was advanced, it lost its economic rationale. The chain of technological improvement broke, and it failed to evolve into a version for cotton.

The fate of the Grand Spinning Wheel is a microcosm of countless inventions in traditional China. Like meteors, they flashed across the sky and vanished from history, swallowed by the involution trap.

The disappearance of such machinery was not an accident; it reflected the same structural impasse that would block industrialization for centuries. Even where intense labor input achieved relative stability in grain yields and allowed for capital accumulation (as in the Yangtze Delta), cheap labor killed the incentive to use machines, while highly volatile markets killed the factories that could house them. Consequently, this capital could not find an industrial outlet. Without the economic conditions to make technological upgrading pay off, even the best machines cannot ignite an Industrial Revolution.

## VII. Contemporary China: A Successful Big Push

To understand the first thirty years of the People's Republic, we must answer one core question: What was the imperative of the revolution?

From the late Qing to the Republican era, traditional agriculture was buckling under population pressure. External shocks from the West broke the old order. Famine, warlords, and invasion pushed Chinese society to the brink of chaos. In this environment of political disorder compounded by geographic volatility, market mechanisms were paralyzed. Industrialization was impossible.

The revolution, therefore, was not merely an ideological choice, but a necessity for survival. Its success lay not just in seizing power, but in reestablishing a viable order out of chaos.

To industrialize, China first had to overcome its age-old vulnerability to geographic volatility. This was a massive undertaking of engineering: a total reconstruction of the nation’s landscape.

From this perspective, the history of the first thirty years takes on a new meaning. It was not a “detour violating comparative advantage” (i.e., prioritizing capital-intensive heavy industry despite abundant labor and scarce capital), but a Big Push that solved the binding problem. For late-developing nations facing systemic underdevelopment, a Big Push, meaning coordinated investment to break industrial bottlenecks, must weather the fundamental challenge of grain volatility and ultimately overcome it through the very power generated by industrialization, or the Big Push will fail to deliver self-sustaining growth.

## 1. The Planned System: A Buffer against Volatility

At the founding of the People's Republic, China was essentially still a pre-industrial agrarian society. Grain production was volatile, hovering precariously around survival levels. It was in this desperate context that China adopted the planned system.

Critics often blame the planned system for the shortage economy. Shortages were undeniable, but this criticism confuses cause with effect: it was not the plan that caused the shortage; it was the shortage, chronic scarcity and volatility, that necessitated the plan.

China had always suffered from shortages—long before the planned system existed. In the past, shortage meant famine; industrial capacity was equally absent. As Mao Zedong famously lamented: "What can we make now? We can make tables and chairs, teacups and teapots, we can grow grain, grind it into flour, and make paper, but we can't make a single car, a single plane, a single tank, or a single tractor."

Yet, industrialization requires capital—and capital must be denominated in a currency that holds real value. But what gives money its value? The value of money is derived not from gold, but from the command over wage goods—essential commodities like food and clothing that people need to survive. As Chen Yun, the architect of China's early economy, insisted, the amount of currency that could be issued was determined by the supply of physical goods, above all, grain and cotton cloth.

**Money holds value only when it is backed by grain.** In a highly volatile environment, limited grain surplus was precarious; a free market would eventually succumb to hoarding, hyperinflation, and collapse.

Thus, here lies the essence of the planned system: in a context where high volatility rendered uncoordinated markets dysfunctional, the state used its limited grain surplus as an anchor for the currency, engineering an insulated industrial sphere as a substitute for the market. Within this safe haven, heavy industry and research sectors could accumulate capital and advance technology, isolated from external turbulence.

The planned system served as a buffer against high volatility, sustaining the nascent industrial base.

## 2. The Silent Agricultural Revolution

In the first thirty years, China completed a deep transformation of nature with one core goal: neutralizing survival risks.

**The Big Push:** centered on 156 key industrial projects backed by the Soviet Union, the Big Push built a complete industrial backbone. But more importantly, it laid the foundation for the agricultural revolution.

**The overlooked battleground:** the most critical yet underappreciated achievement was the systematic engineering in agriculture. To achieve stable yields under the monsoon climate, a thorough re-engineering of the physical landscape was necessary. By focusing on four key needs (water, fertilizer, pesticides, and better seeds), China achieved a silent agricultural revolution:

- **Irrigation and Flood Control:** upon the founding of the People's Republic, a massive struggle to tame rivers began, from the Huai River campaign to the countless reservoirs across the country. By the late 1970s, China had built approximately 80,000 reservoirs, irrigating in drought and draining in flood.
- **Fertilizer:** starting from scratch in the 1950s, China invested heavily in fertilizer plants, importing 13 massive complexes in the 1970s. By 1980, usage reached 12.7 million tons, and by 1990, it exceeded 25 million tons. This surge in the early 1980s meant agriculture finally broke its natural limits: utilizing industrial chemistry to turn fossil fuels into nitrogen for the soil. This was the critical tipping point where per capita output shifted from stable to soaring.
- **Pesticides:** a basic pesticide industry was established to curb pest outbreaks, helping reduce crop losses.
- **Seeds:** breakthroughs in crop breeding (including hybrid rice, improved wheat varieties, and hybrid maize) significantly increased yields while making crops more resistant to pests and unpredictable weather.

The impact of this revolution was transformative. In the first thirty years, while the population nearly doubled (from 540 million to nearly 1 billion), per capita grain output was broadly maintained around 300 kg. Maintaining per capita output despite a population explosion was a significant outcome. By the early 1980s, as these investments matured, grain yields soared. Per capita output climbed steadily, breaking the Malthusian ceiling.

The conventional narrative credits the Household Responsibility System, the return to family farming in the early 1980s, as the decisive cause of the agricultural boom. This view overstates the role of incentives while understating the material preconditions that made them effective. China had practiced household farming for millennia; it had never, by itself, solved the problem. In a highly volatile environment, no incentive mechanism, however well-designed, can overcome the systemic risk that destroys surplus as fast as it is created. What changed by 1980 was not the incentive structure alone, but the physical environment in which it operated. Reservoirs tamed floods and droughts; fertilizer broke natural yield ceilings; improved seeds and pesticides reduced

crop losses. It was this engineered stability that made household incentives productive for the first time in Chinese history.

In contractual terms, the Household Responsibility System was structurally a fixed-rent arrangement: the household owed a fixed quota to the collective and the state, and retained all surplus beyond it. This is precisely the logic of the English tenant farmer paying a fixed rent and claiming the residual. China had known fixed rent for over a millennium. The *iron-rent* of the Yangtze Delta was its most recent incarnation. But as we showed earlier, in a highly volatile environment, fixed rent remained nominal across most of the country: disasters forced repeated abatement, collapsing it back into de facto sharecropping. Genuine rigidity existed only in rare engineered oases like the Dujiangyan irrigation zone, too small to alter the national trajectory. What made the post-1980 fixed quota genuinely rigid was the same factor that made English fixed rent rigid: low output volatility. Engineering had suppressed the variance to the point where the quota could be reliably met year after year, and the household, now a genuine residual claimant, had every incentive to maximize output. The Dujiangyan logic, once confined to a 2,000-square-kilometer oasis, had finally been replicated at a continental scale.

The resulting grain abundance served a function far beyond household wealth: it stabilized the national food supply, anchored the currency, and released hundreds of millions of hands from the land into industry, setting the stage for the virtuous cycle that followed.

The sequencing matters: first, the collective system built the material foundation; then, the restoration of household incentives released the energy that the foundation could now support. This was the deepest institutional change in China's history—a transformation not of political form, but of the baseline institution itself, made possible only by the reconstruction of the material foundation upon which all institutions rest.

## VIII. Contemporary China: Self-sustaining Growth

The planned system had served as a safe haven, nurturing infant industries in an insulated environment. But its capacity was inherently limited: with a weak currency anchor rooted in a limited grain supply, the system could only sustain a narrow division of labor. Once the silent agricultural revolution crossed the material threshold, a well-functioning market could finally take root and thrive. The full-scale industrialization of the last forty years was no longer driven by the state's Big Push, but became self-sustaining once the stable foundation had been built.

### 1. The Virtuous Cycle Begins

The foundations laid during the first thirty years finally enabled a breakthrough in the early 1980s.

First, the state mitigated the effects of geographic volatility. A stable food supply provided a solid anchor for the currency. With food finally abundant and accessible, the rationing system and

strict controls on population movement loosened, and the real costs of labor and commodities became predictable. On one hand, lured by the prospect of industrial profits and tax revenue, investors and local governments mobilized to build factories. On the other hand, hundreds of millions of farmers could finally leave the land and enter factories. This ultimately ignited the explosive industrialization spearheaded by Township and Village Enterprises (TVEs) in the 1980s.

Second, the state slashed the factor costs of production. Railways and roads expanded steadily, supporting large-scale logistics. A robust power grid gradually covered the country, providing more reliable energy than anywhere else in the developing world. This extensive transport and power network allowed the production system to operate at extremely low costs, integrating scattered regional markets into a unified national market.

Thus, we witnessed one of the most rapid and extensive expansion of the division of labor in human history. Hundreds of millions of farmers entered factories to specialize in a single task. Spindles spinning in textile mills, assembly lines humming in electronics factories—everyone did only one thing, and productivity skyrocketed.

By unleashing the Smithian logic of specialization across its vast workforce, China became the “world's factory”. This transformation went beyond cheap labor; it was the deepening of specialization itself—the true source of China's manufacturing power. Compared to the Global South, where industrial systems remain fragmented, China possesses highly integrated supply chains. Compared to the West, where institutional friction and aging infrastructure often constrain efficiency, China possesses substantial scale and structural cost advantages. Ultimately, this all-round strength allowed Chinese manufacturing to reach historically unprecedented levels of efficiency and scale.

More importantly, China has not stopped there. Building on its integrated supply chains, China is spearheading the New Energy Revolution, reinforcing the material foundation of its market.

If the Agricultural Revolution solved the problem of food for humans, the New Energy Revolution is solving the “problem of food for machines”. Breakthroughs in nuclear, solar and wind power, as well as battery storage, mean energy will become cheaper, more stable, and more ubiquitous than ever before. This will further reduce production costs and iron out the risks of global energy volatility.

This is a reinforcing feedback loop: a well-functioning market enables full industrialization, which generates new energy technology, which in turn reinforces the physical foundation for the market. This cycle pushes China toward the next frontier of the industrial revolution.

## 2. Engineering a Well-functioning Market

In analyzing the China model, Professor Justin Yifu Lin proposed the twin concepts of "facilitating state" and "effective market" (which we term "well-functioning market"). He argues that the government should provide infrastructure, coordinate investment, and facilitate industrial

upgrading, while the market allocates resources. This is a profound observation. However, integrating this observation with history reveals a deeper causality.

For late-developing nations, a well-functioning market does not emerge spontaneously. Nor does it grow automatically just by hanging up a free market sign, privatizing assets, and freeing prices. A well-functioning market requires a solid anchor for the currency. This anchor rests on an abundant, stable supply of food and energy, achievable only through systemic engineering.

Therefore, the fundamental logic of the China model is: t**he state engineered the material conditions for a well-functioning market.**

Professor Lin also views the heavy industry drive of the first thirty years as a "detour"—a violation of comparative advantage. In this context, comparative advantage means that China—with abundant labor and scarce capital—should have developed light industry first, rather than jumping into capital-intensive heavy industry. Lin argues that this detour led to inefficiencies and shortages. But if we shift our lens from static efficiency to dynamic evolution, a different rationale emerges.

Why did China insist on defying comparative advantage to develop heavy industry? Beyond national defense, an often-ignored economic reason is that the so-called labor advantage, while theoretically possible, was physically out of reach in that highly volatile environment. A large population does not automatically translate into an available industrial workforce; without a stable food surplus, the massive population was tied to the land, forced to grow their own food just to survive. Even locally organized production, which does not require labor migration, never advanced beyond household handicraft precisely because demand volatility destroyed the stable market that factory production requires. And handicraft could not produce the means to solve the binding problem itself.

To untie this workforce, agriculture had to achieve high and stable yields, and to achieve that, China had to use heavy industry to engineer away the material constraints:

- Without steel and cement, there would be no dams or irrigation networks.
- Without an advanced machine-building industry, it would be impossible to manufacture the high-pressure equipment required for synthetic ammonia (the key to fertilizer).
- Without the chemical industry, there would be no pesticides.
- Without the power industry, there would be no rural electric grid to power the vast network of electric irrigation and drainage stations.

It was this industrial network that first secured the survival floor against droughts and floods, and then progressively drove yields beyond natural limits.

Therefore, the heavy industry strategy, while it appeared to violate comparative advantage through the lens of static efficiency, was actually an inevitable choice grounded in physical constraints. By channeling industrial capacity into agriculture, the state systematically

neutralized the effect of geographic volatility and stabilized the price system. Once this material threshold was crossed, the comparative advantage in labor finally became an economic reality.

This is the root of the China model: while a well-functioning market may coordinate economic activity, it cannot bring itself into being. In a highly volatile environment, the state cannot just “facilitate” the market; it must engineer the material stability that allows a well-functioning market to exist in the first place. First, the state reshapes the material foundation and tames volatility. Then, and only then, can the market mechanism take root and thrive.

To truly understand the China model is to understand why some have argued that Deng Xiaoping should not be called the "chief architect" of reform and opening up, but rather the "chief approver": the most effective institutional reforms were not designed from above, but grew from below once the material foundation was in place.

## IX. Governance: Preservation and Transformation

The difference between Chinese and American governance is often summarized as engineer governance versus lawyer governance. Yet, this distinction runs deeper than style. It traces back to the two origins introduced earlier: the law from Sinai and the engineering from Tu. These represent the fundamental divide between preservation and transformation.

This historical logic defines the state's primary mission. In a stable environment, the state functions primarily as a referee of established rights, preserving a social order that nature has already made secure. However, in a highly volatile environment, the state’s first mandate is to mobilize the people to engineer the very stability that nature denied. Here, the priority is not settling disputes, but actively creating the material conditions for survival.

The Western model stems from the English experience. In that providential oasis, grain output was stable, society was peaceful, and external threats were minimal. The government did not need to battle highly volatile harvests, civil unrest, or invading armies. In this context, upholding established rights was all that was needed.

Thus, the West developed the ethic of procedural justice. Its core is preservation: the system prioritizes the integrity of established rules over rapid intervention, a viable strategy in an environment where existential crises are rare. In practice, this governance style is “muddling through”—a low-cost strategy that only a safe world can afford. Even if the government is inefficient, society runs fine on inertia.

The United States has pushed this style to its extreme. As a “supersized England”, the US also won the geographic lottery: stable agriculture, secure borders, and vast oceans separating it from old-world strife. Living in this comfort zone for centuries, the United States has developed a highly formalized governance structure where due process is paramount. While this ensures a deep reliance on the rule of law and the protection of established rights and boundaries, it also leads to a more litigious environment where legal deliberation often precedes real-world action.

Without the immediate pressure of an existential threat, adherence to procedure becomes the primary mechanism for maintaining social consensus. In such a system, the necessity for broad agreement often leads to what has been termed a vetocracy, where the numerous veto points slow the pace of resource mobilization. This is not a failure of will, but a structural trade-off: a system designed to protect many different interests in a stable environment naturally prioritizes procedural integrity over the speed of action.

In contrast, China faces a highly volatile environment. Amidst floods, droughts, and the social upheaval they triggered, no eloquence can substitute for action; if the state cannot tame the waters or sustain the population, it loses its "Mandate of Heaven." Here, legitimacy is derived not from the process, but from the tangible outcome of survival. Consequently, the delivery of results becomes the primary political ethic. The state cannot be a mere referee; it must be an active builder, judged by its concrete record of development. In this context, order is not a pre-existing condition to be protected, but a physical outcome that must be engineered.

The engineer's focus on problem-solving thus establishes the physical foundation required for legal certainty to take root, allowing rights to be meaningfully defined and effectively enforced. As we saw with England, stable output anchors prices, lowers enforcement costs, and reduces the transaction costs of social coordination. In essence, China's Big Push was precisely a massive effort to lower systemic transaction costs that nature had initially set too high. Its success in agricultural de-risking allowed China to engineer what England had received as a providential gift: the material foundation for legal certainty and self-sustaining growth.

## X. The Soviet Tragedy: Big Push that Failed to De-risk

The Soviet Union provides a telling contrast; its failure illuminates China's success.

As the first socialist state, the Soviet Union served as the template for China's system. It demonstrated powerful organizational capacity and launched the original Big Push through central planning. But the result was tragic: until its dissolution, it never solved its agricultural crisis. Both geographic constraints and strategic failures were to blame.

**First, the geographic constraints:** Although vast, the Soviet Union's core agricultural zones were confined to high latitudes—heat was insufficient, growing seasons were short. Worse, the harsh continental climate meant erratic rainfall, bringing recurrent drought. These constraints kept grain output both low and volatile: in the 1980s, grain output swung violently between 150 million and 210 million tons, with frequent crop failures.

**Second, the strategic failures:** The planned system allowed the Soviet Union to achieve world-renowned feats (satellites in space, waves of tanks), yet agriculture, the survival base, remained fragile. The Soviet Union launched massive land reclamation campaigns, trying to force a breakthrough in yields. But constrained by the harsh climate, these efforts often yielded meager returns despite massive investment. Even more damaging, unlike China's systematic approach, the Soviet Union's agricultural investment was structurally unbalanced.

- **Neglect of Infrastructure:** Extensive use of fertilizer was limited by insufficient irrigation.
- **Logistical Failure:** Outdated storage and transport systems led to post-harvest losses of up to 20%, tens of millions of tons rotting in fields or in transit.

**The consequence:** Unable to provide stable and abundant grain as an anchor for the currency, the Soviet Union remained confined to the limited sphere of the planned system, failing to create the vast, integrated well-functioning market that China eventually engineered. This narrow division of labor left Soviet industry inefficient and its goods uncompetitive on the global market.

With chronic food shortages and uncompetitive industrial goods, the Soviet Union, unwilling to bear the political cost of continued austerity, embarked on a path of high-risk external dependence: exporting oil to import grain. Grain imports climbed year by year from the 1970s, reaching 30–40 million tons annually by the 1980s.

But oil prices are highly volatile. This oil-for-grain dependency became the Achilles' heel of Soviet finance. In 1985–1986, international oil prices collapsed from around $30 to below $10 and hovered at low levels for years. Revenue plummeted, but the rigid demand for grain remained. Coupled with the arms race and the war in Afghanistan, the fiscal deficit exploded. This long-term fiscal bleeding eventually triggered the dissolution of the Soviet Union.

The tragedy of the Soviet Union shows that state capacity to mobilize resources is a necessary condition for Big Push miracles, but not a sufficient one. Success requires that the Big Push solve the binding problem: systemic agricultural de-risking. This demands respect for physical limits and strategic patience. The Soviet failure lay not just in geography, but in strategy. When agricultural breakthroughs stalled, it did not persist in the arduous transformation of its agricultural base. Instead, it prioritized geopolitical ambitions, channeling volatile oil revenues into welfare and an arms race. This unsustainable dependency came at a heavy price: the Soviet Union collapsed, and wealth built over generations was dissipated.

China's luck lay in its warmer climate; but more importantly, China showed the strategic persistence that the Soviet Union lacked. Even in the hardest times, China never gave up on long-term investment in water infrastructure and fertilizer production, and built an efficient grain transportation and reserve system, securing its own food supply. Avoiding both external dependency and mortgaging the future is the key to China's resilience.

Not every nation can fully secure its own agricultural base. Those that cannot must rely on international markets or powerful allies. The former offers no guarantee of stability, the latter demands a price in sovereignty. China's achievement was to secure the staple grain supply that anchors its domestic market, even as it remains reliant on imports for soybeans and other feed crops.

Today, Russia, the Soviet Union's principal successor, has largely overcome its agricultural crisis, aided by technological progress, market reforms, and a warming climate that has extended growing seasons. Yet for much of the Global South, the challenge remains far more severe.

Crossing the material threshold will demand far greater technological and resource commitments than have so far been mobilized.

## A Supplementary Case: The Irish Path

To further refine our understanding of the geographic lottery, we can examine the diverse paths of European industrialization. In the "Goldilocks zone" of southeastern England, the Low Countries, and northern France, favorable endowments (high mean) converged with stability (low variance). Throughout most of the continent, however, these two variables move in lockstep: moving east or south from this core, yields typically decline just as volatility rises. This synchronized shift across Europe's geography obscures the distinct contribution of each factor, making the case of Ireland, where these dimensions were decoupled, particularly revealing.

Table 1: Endowments, geographic stability, and industrialization outcomes across the West, Russia, and China.

| | Mean-Variance (19th Century) | | | Baseline Institution | Threshold |
|---|---|---|---|---|---|
| Regions | Transport Connectivity | Grain Yield | Geographic Stability | Land tenure | Industrial Takeoff |
| Yangzi Delta China | 5 | 4-5 | 2 | sharecropping | 1950 (Big Push) |
| Ireland North | 4 | 3 | 5 | fixed-rent | 1840 |
| Ireland South | 3 | 3 | 5 | fixed-rent | 1960 |
| England | 5 | 3-4 | 5 | fixed-rent | 1780 |
| American North | 4 | 2-3 | 4 | fixed-rent | 1830 |
| France North | 4 | 3-4 | 5 | fixed-rent | 1830 |
| Netherlands | 5 | 3-4 | 5 | fixed-rent | 1860 |
| Belgium | 5 | 4-5 | 5 | fixed-rent | 1820 |
| Germany North | 4 | 2-3 | 3-4 | fixed-rent | 1840 |
| Germany South | 3 | 3-4 | 3-4 | fixed-rent | 1880 |
| France Central | 3 | 2-3 | 3-4 | complex | 1890 |
| Italy North | 5 | 3-4 | 3-4 | complex | 1890 |
| Spain Industrial | 3-4 | 2-3 | 3-4 | complex | 1840 |
| France South | 3 | 2-3 | 2 | sharecropping | 1950 |
| Italy South | 3 | 2 | 2 | sharecropping | 1950 |
| Spain Agricultural | 2-3 | 1-2 | 2 | sharecropping | 1950 |
| Greece | 3 | 1-2 | 2 | sharecropping | 1950 |
| Ukraine (USSR) | 3-4 | 3 | 3 | complex | 1930 (Big Push) |
| Russia (USSR) | 1-2 | 1-2 | 1-2 | labor-rent | 1930 (Big Push) |

Note: Transport Connectivity, Grain Yield, and Geographic Stability are rated on a relative scale from 5 (Highest) to 1 (Lowest), intended for cross-regional comparison rather than absolute measurement. Ranges (e.g., 2-3, 3-4) are allowed to reflect intra-regional and temporal variation. Geographic Stability refers to the interannual variance of grain yields under prevailing climate. "Complex" land tenure indicates a mix of fixed-rent and sharecropping.

## 1. The Phenomenon of "Stable Mediocrity"

Historically, Ireland presents a unique path: it was a region that possessed geographic stability, yet struggled with the physical hurdles of transport and grain yield. (by ‘stability’ we mean low interannual variance in grain yields under a maritime climate; this does not deny the potato-specific vulnerability that produced the Great Famine.)

While Ireland was often characterized as "poor" relative to the north-west European core, its per capita GDP remained remarkably steady, about one-half to two-thirds of Britain across most of 19th–20th centuries. Within the mean-variance framework, this represents "Stable Mediocrity": the low variance (stability) ensured that Ireland stayed in the economic race over centuries even when faced with external suppression, while its lower mean (mediocre grain yields, hostile terrain, and poor internal connectivity) constrained its development, keeping it on a lower plateau compared to its neighbor across the Irish Sea.

## 2. The North-South Divergence

The internal divergence between the North (Belfast) and the South (Dublin/Cork) during the 19th century illustrates how physical landscapes and institutional niches dictate the material threshold of industrialization, even when regions share the same geographic stability.

Ireland's geographic stability supported the widespread adoption of fixed-rent tenancy, consistent with the pattern observed across stable regions. However, under colonial rule, landlords could for much of the 19th century unilaterally raise rents and evict tenants at will, weakening the developmental function of fixed rent across the island. In the North, this damage was partially offset by colonial political arrangements that afforded Protestant tenants greater security, and by the linen industry, which provided an alternative channel for local capital accumulation. In the South, no such offset existed.

- **The Northern: Integrated Advantages.** Northern grain agriculture was comparable to—and in many districts inferior to—the South’s wheat belt, but Belfast possessed a "friendly" physical landscape. Its navigable river systems and solid terrain allowed for the low-cost integration of its hinterland with its ports; this physical openness allowed the North to compensate for its mediocre local yields by accessing external resources. Proximity to British coal and grain markets provided a "virtual" abundance. Crucially, Britain permitted the linen industry to thrive in Ireland as a non-competing sector. In the North, cool, wet conditions and the heavy, often poorly drained drumlin soils made wheat relatively unattractive, while Ulster’s climate, water resources and smallholder system supported flax cultivation and the rise of linen. Northern capital found a viable industrial exit, allowing a spontaneous industrial takeoff by 1840.
- **The Southern: Physical Vulnerability and Institutional Shackle.** The South faced a far more challenging physical base. The high-rain environment produced a landscape of lush grass but treacherous mud—fostering a thriving livestock sector while severely limiting

grain yields and internal connectivity.

- The Cost of Physical Weakness: The deeper constraint was not middling yields but the small extent of wheat land and, therefore, a limited marketable grain surplus. Southern Ireland's wheat yields were middling by European standards, yet a crop mix dominated by oats and potatoes, combined with bog-ridden, fragmented interiors and high transport costs, curtailed tradable output and the fiscal-military base of the state. This physical reality amplified strategic vulnerability and helped make durable subjugation (often described as 'internal colonization') easier. If the South had possessed a landscape more favorable to grain and transport, it might have lacked its world-class pastoral wealth, but it would have gained the internal cohesion necessary to defend its sovereignty.
- The Institutional Shackle: Once conquered, the South faced a second constraint: Britain's 1699 Woollens Act deliberately constrained Ireland's woollen manufactures by banning direct exports to foreign markets and subjecting shipments to Britain to heavy duties, while leaving exports of raw wool (and yarn) to Britain legal, disproportionately hurting the southern woollen regions. Though formal restrictions eased after 1800, the damage was done: the South had lost the critical window for building industrial capacity, leaving it unable to compete when the market finally opened. The net effect was to limit the highest-value segment of the South's most natural industry and tilt investment toward pastoral provisions (butter, salted beef) rather than deepening a woollen value chain; southern ports increasingly functioned as raw-material conduits, and wealth drained to Britain.

## 3. Revolution and Redirection in the South

The 1922 independence revolution created the policy autonomy to progressively reduce the drain and redirect surplus domestically. By gaining sovereignty, southern Ireland (initially named the "Irish Free State", today the "Republic of Ireland") did not suddenly become rich; rather, it finally gained the power to keep its accumulated surplus at home.

With the surplus now retained at home, providing the foundation for credit and investment, the independent government's first priority was to industrialize what the land already produced reliably. The fiscal base this generated then funded the infrastructure projects that expanded and deepened the domestic market:

- **Industrialize the Surplus:** Ireland did not leap into an economic miracle; it first built the "stomach" to digest its own wealth by industrializing its existing stable production. Agro-processing had deep roots predating independence: dairy cooperatives had spread since the 1890s, and brewing and distilling were already major industries. But under colonial rule, much of the profit drained to Britain. Independence allowed this surplus to be retained and reinvested domestically. From the 1920s onward, the state actively expanded this base: meatpacking scaled up, dairy cooperatives consolidated, and world-class brewing and distilling transformed what were once "live exports" into high-value processed goods. Because these industries relied on a steady supply of raw

materials, a requirement that Ireland's stability could reliably meet, they provided the foundational industrial coordination and capital accumulation that allowed the nation to cross the material threshold and eventually outpace its former master.

- **Securing the Energy Anchor:** Starting in the 1920s with the Shannon Scheme, a massive hydroelectric project, Ireland began to engineer its own energy sovereignty. This was followed by systematic peat development from the 1930s through the 1960s. The government mechanized peat extraction and commissioned peat-fired power stations, expanding the domestic fuel base. This partly reduced reliance on imported coal and provided a measure of energy security to support industrial growth. Recognizing that the industrial growth of the 1960s and 1970s remained fragile due to a heavy reliance on volatile oil imports, the state launched a decisive wave of energy de-risking in the 1980s. The commissioning of large coal-fired baseload capacity and the arrival of domestic offshore natural gas, later reinforced by a gas interconnector to Britain, helped anchor the power system in more stable, predictable sources. By the 1990s, Ireland had largely reduced oil exposure in power generation, enhancing the infrastructure foundation required for sustained industrial growth.
- **Integrating the Interior:** The new state inherited a rail network designed to extract resources for Britain: a radial system centered on Dublin and the ports, with weak internal connections. Repurposing this colonial infrastructure took decades. From the 1950s, the focus shifted to road development. Uneconomic rail lines were closed, and road transport, especially motor freight, expanded rapidly. The real transformation came with European funds: from the 1970s, EC regional funds supported highway improvements, and after 1988, EU structural funds enabled large-scale motorway construction. From the 1990s onward, the interior was gradually connected by modern roads, transforming a once-fragmented landscape into an increasingly integrated domestic market.

## 4. The Modern Leap

Ireland (the Republic of Ireland) did not invent prosperity, it simply removed the physical and institutional dampeners that had long suppressed it. This patient reconstruction of the material base ensured that when the global opportunity, driven by deepening European integration and the surge of footloose high-tech capital, arrived in the 1990s, Ireland’s foundation was finally hard enough to support an economic miracle.

Ireland’s journey proves that endowments are not destiny, but constraints to be overcome. Unlike nations trapped in highly volatile environments that require a Big Push just to kick-start development, Ireland possessed the stability for persistent accumulation: incremental engineering within a well-functioning market, where each project could build on the last and generate returns over time. By leveraging a stable fiscal base to mobilize domestic and external capital, Ireland transformed itself from a "stable but mediocre" pastoral society into a global industrial frontier. It is the story of a nation that used its "geographic lottery ticket" of stability to rise from subjugation to prosperity.

## Policy Implications: A Successful Big Push for the Global South

The perspective of geographic volatility complements rather than replaces the established perspective of geographic endowments. While adverse geographic endowments (low mean) explain the poverty traps of certain regions, it is environmental volatility (high variance) that accounts for the stagnation traps of others.

The mean-variance framework synthesizes the perspectives of endowment and volatility; in reality, late-developing nations facing systemic underdevelopment typically suffer from both constraints of low mean and high variance simultaneously. Thinking back to the development stories of East and West, we can draw a crucial lesson for the present: underdevelopment is not, at root, a product of cultural traits or bad governance, but a natural result of these severe geographic constraints.

- **West Asia and North Africa:** an inherently water-scarce region. In the modern era of population explosion, this is no longer a problem that can be solved by building a few reservoirs; it is an absolute scarcity.
- **Sub-Saharan Africa:** a region facing multiple geographic curses. Tropical diseases weaken the workforce; rainforest and savanna climates make the threshold for agricultural development exceptionally high; and the lack of navigable rivers makes inland transport prohibitively expensive.
- **India:** another case of high volatility. India's agricultural conditions are far inferior to China's. Its rainfall relies heavily on the erratic South Asian monsoon, and the intense heat fosters pest outbreaks. This makes India's grain output highly volatile, causing a severe governance paradox. Volatile yields make the tax base difficult to assess and collect; insufficient tax revenue means insufficient bureaucrats and infrastructure. Historically this prevented India from establishing a strong centralized state. While Western electoral democracy was transplanted, in a society fragmented by caste, religion, and region, it inevitably became an arena where competing interests must be reconciled, continuing, in a new form, the long historical challenge of governing a diverse subcontinent. Aided by the Green Revolution and its own industrial investments, India has moved beyond the worst of its poverty trap, but the path forward remains slowed by the sheer difficulty of forging consensus across its vast and diverse society.

Western nations, accustomed to the luxury of a guaranteed harvest, assume that as long as Western institutions are copied, prosperity will follow. However, this assumption fails to account for the material constraints: in highly volatile environments, institutional blueprints cannot be implemented without first securing a stable physical foundation.

As economist Jeffrey Sachs has pointed out, for nations trapped in poverty, standard market prescriptions simply do not work; they need a Big Push—massive coordinated investment in infrastructure and agricultural inputs. This diagnosis has struggled to gain traction against alternative approaches that focus on marginal interventions within the existing environment

rather than transforming the environment itself. Microfinance seeks to unlock individual entrepreneurship through small loans; randomized controlled trials test the impact of targeted provisions such as bed nets or fertilizer subsidies; property titling aims to convert informal assets into collateral. Each addresses a real constraint at the individual level. But from a systemic perspective, they share a common blind spot: none of them alters the systemic volatility that periodically destroys the gains they enable. In an environment where a single drought can wipe out a season's earnings, a small loan cannot build a durable enterprise; a bag of fertilizer cannot compensate for the absence of irrigation; and a property title cannot hold its value when the harvest fails. They may bring temporary relief, but in a systemically volatile environment, such gains are written in sand, erased by the next shock before they can accumulate. These are not failed ideas but incomplete ones: they operate within the constraints that only systemic engineering can remove.

Even when the need for a Big Push is recognized, many of these nations are caught in the governance paradox imposed by severe geographic constraints: the very conditions that demand a Big Push also undermine the state capacity to initiate and sustain one; both launching a Big Push and sustaining it until self-sustaining growth is achieved become immense challenges. Thus, for these nations, external partnership can play a crucial role in breaking the trap.

China's Belt and Road Initiative (BRI), along with its agricultural cooperation programs across the Global South, resonates with Sachs's "clinical economics." By investing in transport and energy infrastructure, and by transferring agricultural technology such as hybrid rice and irrigation techniques, China is not just offering temporary relief; it is helping build the material foundation that must precede a well-functioning market. Though this may appear to have little to do with institutions, it is precisely what transforms the baseline institution, and that is the institutional prescription that truly works.

Yet this is only the very beginning. A successful Big Push for the Global South will require going beyond current efforts to address the deeper constraints of food and water security, with far more coordinated planning and far greater resource commitment than what has been mobilized so far. Such commitment is justified on both moral and practical grounds: as more nations overcome geographic constraints and build well-functioning markets, they join as new participants who expand the global market and generate gains for all. China engages with the Global South as a trade and infrastructure partner because its own rise was built on a proven sequence: build the foundation first, and prosperity follows.

## Conclusion: The Renewal of Civilization

The story of modernization is not the triumph of one culture over another, but a testament to humanity's capacity to master its own material environment. It proves that poverty and stagnation are not destinies dictated by the physical landscape. Even in the most volatile environments, the material threshold of industrialization can be crossed through a Big Push solving the binding problem: systemic agricultural de-risking. By constructing a stable material foundation, human society can finally break the shackles of geographic constraints.

For millennia, civilizations were trapped in scarcity and unpredictability. Because the material world was volatile, surpluses could not be sustained, and competition for limited resources often turned violent, leading to a global order anchored in plunder and war. Systemic engineering is breaking this cycle. When humanity uses technology to stabilize the supply of food and energy, human cooperation shifts from a struggle for survival to a positive-sum game of continuous accumulation. This creates the first real opportunity to move beyond the defensive logic of the old era. The focus of governance shifts from prescribing institutional forms to constructing material foundations, and the world order is moving toward a new era where building replaces snatching.

This journey echoes the ancient narrative of Jacob wrestling with the divine through the long night until the break of dawn. In the pre-modern era, humanity was "Jacob", a wanderer at the mercy of an unpredictable Providence, struggling against the chaos of a volatile nature. By achieving material predictability through engineering, we are realizing the deeper meaning of the name "Israel": those who wrestle with fate and prevail.

China's experience is but one manifestation of this universal trajectory. Humanity is moving toward a shared future where engineered stability replaces geographic chance as the bedrock of order. We are building a world where human dignity is no longer a providential luxury, but an engineered baseline secured for all.